



Aalto University

On nonstationary subspace analysis for multivariate random fields

Joint work with Perttu Saarela, Sandra De Iaco and Klaus Nordhausen

Jaakko Pere

LinStat 2026

Agenda of the Presentation

Motivation and model definition

Population version of the estimator

Sample version of the estimator

Table of Contents

Motivation and model definition

Population version of the estimator

Sample version of the estimator

Linear blind source separation for random fields

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathbb{R}^p$. Consider a model

$$\begin{pmatrix} x_1(\mathbf{u}) \\ \vdots \\ x_d(\mathbf{u}) \end{pmatrix} = x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \mathbf{A} \begin{pmatrix} z_1(\mathbf{u}) \\ \vdots \\ z_d(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

Linear blind source separation for random fields

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathbb{R}^p$. Consider a model

$$\begin{pmatrix} x_1(\mathbf{u}) \\ \vdots \\ x_d(\mathbf{u}) \end{pmatrix} = x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \mathbf{A} \begin{pmatrix} z_1(\mathbf{u}) \\ \vdots \\ z_d(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

$\Rightarrow$ We observe x but z has simpler dependence structure. The goal is to extract $\mathbf{A}$ or z .

Linear blind source separation for random fields

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathbb{R}^p$. Consider a model

$$\begin{pmatrix} x_1(\mathbf{u}) \\ \vdots \\ x_d(\mathbf{u}) \end{pmatrix} = x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \mathbf{A} \begin{pmatrix} z_1(\mathbf{u}) \\ \vdots \\ z_d(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

$\Rightarrow$ We observe x but z has simpler dependence structure. The goal is to extract $\mathbf{A}$ or z .

- Components of $z(\mathbf{u})$ are uncorrelated and *stationary* (Bachoc, Genton, et al. 2020).
- Components of $z(\mathbf{u})$ are independent and *nonstationary* (Bachoc, Muehlmann, et al. 2024).

Our model

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathcal{U} \subset \mathbb{R}^p$. Consider the model

$$x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \begin{pmatrix} \mathbf{A}_\eta & \mathbf{A}_s \end{pmatrix} \begin{pmatrix} \eta(\mathbf{u}) \\ s(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

■ $\mathbf{A}_\eta \in \mathbb{R}^{d \times q}$ and $\mathbf{A}_s \in \mathbb{R}^{d \times (d-q)}$.

Our model

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathcal{U} \subset \mathbb{R}^p$. Consider the model

$$x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \begin{pmatrix} \mathbf{A}_\eta & \mathbf{A}_s \end{pmatrix} \begin{pmatrix} \eta(\mathbf{u}) \\ s(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

- $\mathbf{A}_\eta \in \mathbb{R}^{d \times q}$ and $\mathbf{A}_s \in \mathbb{R}^{d \times (d-q)}$.
- Components of q -dimensional field $\eta(\mathbf{u})$ are *nonstationary*.
- Components of $(d - q)$ -dimensional field $s(\mathbf{u})$ are *stationary*.
- The random fields η and s are independent.

For simplicity of the presentation, we only consider nonstationarity in the mean.

Table of Contents

Motivation and model definition

Population version of the estimator

Sample version of the estimator

Local and global statistics (1)

Divide the domain $\mathcal{U} \subset \mathbb{R}^p$ in blocks $\mathcal{S}_j$. Sample in block $\mathcal{S}_j$ is S_j .

- Number of blocks is k . Index blocks by $j \in \{1, \dots, k\}$.
- Sample size in each block is n . Index observations in a block by $i \in \{1, \dots, n\}$.

Local and global statistics (1)

Divide the domain $\mathcal{U} \subset \mathbb{R}^p$ in blocks $\mathcal{S}_j$. Sample in block $\mathcal{S}_j$ is S_j .

- Number of blocks is k . Index blocks by $j \in \{1, \dots, k\}$.
- Sample size in each block is n . Index observations in a block by $i \in \{1, \dots, n\}$.
- We denote that S_0 is the total sample and n_0 is the total sample size.

Local and global statistics (1)

Divide the domain $\mathcal{U} \subset \mathbb{R}^p$ in blocks $\mathcal{S}_j$. Sample in block $\mathcal{S}_j$ is S_j .

- Number of blocks is k . Index blocks by $j \in \{1, \dots, k\}$.
- Sample size in each block is n . Index observations in a block by $i \in \{1, \dots, n\}$.
- We denote that S_0 is the total sample and n_0 is the total sample size.
- Idea for estimation
 - $\Rightarrow$ Compare local statistics to global statistics.

Local and global statistics (2)

Define the following local (and global) statistics

$$\mathbf{m}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E} (x(\mathbf{u})) ,$$

$$\text{Cov}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E} ((x(\mathbf{u}) - m_j(x)) (x(\mathbf{u}) - m_j(x))^{\top})$$

For identifiability assume...

$$\text{Cov}(s(\mathbf{u})) = \mathbf{I}, m_0(\eta) = \mathbf{0} \text{ and } \text{Cov}_0(\eta) = \mathbf{I}.$$

For identifiability assume...

$$\text{Cov}(s(\mathbf{u})) = \mathbf{I}, m_0(\eta) = \mathbf{0} \text{ and } \text{Cov}_0(\eta) = \mathbf{I}.$$

“Location” and “covariance” fixed: It follows that $m_0(z) = \mathbf{0}$ and $\text{Cov}_0(z) = \mathbf{I}$.

Scatter based on blocks

Define

$$M_{k,n}(x) = \frac{1}{k} \sum_{j=1}^k (\mathbf{m}_j(x) - \mathbf{m}_0(x)) (\mathbf{m}_j(x) - \mathbf{m}_0(x))^{\top}$$

Scatter based on blocks

Define

$$M_{k,n}(x) = \frac{1}{k} \sum_{j=1}^k (\mathbf{m}_j(x) - \mathbf{m}_0(x)) (\mathbf{m}_j(x) - \mathbf{m}_0(x))^{\top}$$

Idea:

- For stationary s we have $M_{k,n}(s) = \mathbf{0}$.
- For nonstationary η we have $M_{k,n}(\eta) \neq \mathbf{0}$ is symmetric positive definite.

Extracting the unmixing matrix

1. Whiten the field: $\tilde{\mathbf{x}}(\mathbf{u}, t) = \mathbf{W}(\mathbf{x}(\mathbf{u}, t) - m_0(\mathbf{x}))$, where $\mathbf{W}\text{Cov}_0(\mathbf{x})\mathbf{W}^\top = \mathbf{I}$.

Extracting the unmixing matrix

1. Whiten the field: $\tilde{\mathbf{x}}(\mathbf{u}, t) = \mathbf{W}(\mathbf{x}(\mathbf{u}, t) - m_0(\mathbf{x}))$, where $\mathbf{W}\text{Cov}_0(\mathbf{x})\mathbf{W}^\top = \mathbf{I}$.
2. Diagonalize: Let $\mathbf{V}_{k,n} \in \mathbb{R}^{d \times d}$ be a matrix such that

$$\mathbf{V}_{k,n}\mathbf{V}_{k,n}^\top = \mathbf{I} \quad \text{and} \quad \mathbf{V}_{k,n}^\top M_{k,n}(\tilde{\mathbf{x}})\mathbf{V}_{k,n} = \mathbf{\Lambda}_{k,n},$$

where $\mathbf{\Lambda}_{k,n} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with diagonal elements in nonincreasing order.

Extracting the unmixing matrix

1. Whiten the field: $\tilde{\mathbf{x}}(\mathbf{u}, t) = \mathbf{W}(\mathbf{x}(\mathbf{u}, t) - m_0(\mathbf{x}))$, where $\mathbf{W}\text{Cov}_0(\mathbf{x})\mathbf{W}^\top = \mathbf{I}$.
2. Diagonalize: Let $\mathbf{V}_{k,n} \in \mathbb{R}^{d \times d}$ be a matrix such that

$$\mathbf{V}_{k,n}\mathbf{V}_{k,n}^\top = \mathbf{I} \quad \text{and} \quad \mathbf{V}_{k,n}^\top M_{k,n}(\tilde{\mathbf{x}})\mathbf{V}_{k,n} = \mathbf{\Lambda}_{k,n},$$

where $\mathbf{\Lambda}_{k,n} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with diagonal elements in nonincreasing order.

3. Transform back: an unmixing matrix $\mathbf{\Omega}_{k,n} \in \mathbb{R}^{d \times d}$ based on $M_{k,n}$ is defined as $\mathbf{\Omega}_{k,n} = \mathbf{V}_{k,n}^\top \mathbf{W}$.

Identifiability: definition

The unmixing problem based on $M_{k,n}$ is identifiable if for every matrix $\Omega_{k,n}$ satisfying definition of above slide there exist invertible matrices $\mathbf{B}_\eta \in \mathbb{R}^{q \times q}$ and $\mathbf{B}_s \in \mathbb{R}^{(d-q) \times (d-q)}$ such that

$$\Omega_{k,n} = \begin{pmatrix} \Omega_\eta \\ \Omega_s \end{pmatrix} = \begin{pmatrix} \mathbf{B}_\eta & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_s \end{pmatrix} \mathbf{A}^{-1}, \quad \Omega_\eta \in \mathbb{R}^{q \times d}, \quad \Omega_s \in \mathbb{R}^{(d-q) \times d}.$$

Identifiability: meaning

We obtain “mixed” stationary and nonstationary signals:

- $\Omega_{\eta}x(\mathbf{u}) = \mathbf{B}_{\eta}\eta(\mathbf{u})$

- $\Omega_s x(\mathbf{u}) = \mathbf{B}_s s(\mathbf{u})$

Identifiability: meaning

We obtain “mixed” stationary and nonstationary signals:

- $\Omega_{\eta}x(\mathbf{u}) = \mathbf{B}_{\eta}\eta(\mathbf{u})$
- $\Omega_s x(\mathbf{u}) = \mathbf{B}_s s(\mathbf{u})$

We can choose $\mathbf{B}_{\eta}$ and $\mathbf{B}_s$ to be orthogonal:

- $\text{Cov}(\mathbf{B}_s s(\mathbf{u})) = \mathbf{I},$
- For example, under $\text{Cov}(\eta(\mathbf{u})) = \sigma^2(\mathbf{u})\mathbf{I},$ we have that $\text{Cov}(\mathbf{B}_{\eta}\eta(\mathbf{u})) = \sigma^2(\mathbf{u})\mathbf{I}.$

Identifiability: condition

- Let $\lambda_{\eta}^{(\min)}$ be the smallest eigenvalue of $M_{k,n}(\eta)$
- Let $\lambda_s^{(\max)}$ the largest eigenvalue of $M_{k,n}(s)$.

Identifiability: condition

- Let $\lambda_{\eta}^{(\min)}$ be the smallest eigenvalue of $M_{k,n}(\eta)$
- Let $\lambda_s^{(\max)}$ the largest eigenvalue of $M_{k,n}(s)$.
- Then unmixing problem based on $M_{k,n}$ is identifiable if and only if $\lambda_{\eta}^{(\min)} > \lambda_s^{(\max)}$.

Identifiability: condition

- Let $\lambda_{\eta}^{(\min)}$ be the smallest eigenvalue of $M_{k,n}(\eta)$
- Let $\lambda_s^{(\max)}$ the largest eigenvalue of $M_{k,n}(s)$.
- Then unmixing problem based on $M_{k,n}$ is identifiable if and only if $\lambda_{\eta}^{(\min)} > \lambda_s^{(\max)}$.

Meaning: all the components of η are nonstationary in the mean.

Table of Contents

Motivation and model definition

Population version of the estimator

Sample version of the estimator

Above we focused on scatter based on $m_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E}(x(\mathbf{u}))$.

$\Rightarrow$ In practice the expected value needs to be dropped.

Above we focused on scatter based on $m_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E}(x(\mathbf{u}))$.

$\Rightarrow$ In practice the expected value needs to be dropped.

Define,

$$\hat{m}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} x(\mathbf{u}, t),$$

$$\widehat{\text{Cov}}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} (x(\mathbf{u}) - \hat{m}_j(x)) (x(\mathbf{u}) - \hat{m}_j(x))^{\top},$$

Above we focused on scatter based on $m_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E}(x(\mathbf{u}))$.

$\Rightarrow$ In practice the expected value needs to be dropped.

Define,

$$\hat{m}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} x(\mathbf{u}, t),$$

$$\widehat{\text{Cov}}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} (x(\mathbf{u}) - \hat{m}_j(x)) (x(\mathbf{u}) - \hat{m}_j(x))^{\top},$$

and

$$\hat{M}_{k,n}(x) = \frac{1}{k} \sum_{j=1}^k (\hat{m}_j(x) - \hat{m}_0(x)) (\hat{m}_j(x) - \hat{m}_0(x))^{\top}$$

Our asymptotic framework

- Sample size per block and number of blocks: $n \rightarrow \infty$ and $k \rightarrow \infty$ ($n_0 = nk \rightarrow \infty$).

Our asymptotic framework

- Sample size per block and number of blocks: $n \rightarrow \infty$ and $k \rightarrow \infty$ ($n_0 = nk \rightarrow \infty$).
- Increasing domain: $\text{Leb}(\mathcal{U}) = \text{Leb}(\mathcal{U}_{n_0}) \rightarrow \infty$, as $n_0 \rightarrow \infty$.

Our asymptotic framework

- Sample size per block and number of blocks: $n \rightarrow \infty$ and $k \rightarrow \infty$ ($n_0 = nk \rightarrow \infty$).
- Increasing domain: $\text{Leb}(\mathcal{U}) = \text{Leb}(\mathcal{U}_{n_0}) \rightarrow \infty$, as $n_0 \rightarrow \infty$.
- Suppose that the smallest eigenvalue $\lambda_{\eta,k,n}^{(min)}$ of $M_{k,n}(\eta)$ satisfies

$$\liminf_{k,n \rightarrow \infty} \lambda_{\eta,k,n}^{(min)} \geq c, \quad c > 0.$$

Asymptotic identifiability

If we have

$$\left\| \hat{M}_{k,n}(z^{st}) - M_{k,n}(z) \right\| = O_{\mathbb{P}}(r_{k,n}),$$

and that

$$\left\| \widehat{\text{Cov}}_0(\mathbf{z}) - \mathbf{I} \right\| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n_0}}\right),$$

where $1/\sqrt{n_0} = O(r_{k,n})$, then there exists a sequence of random invertible matrices $\hat{\mathbf{B}}_{\eta}^{(k,n)} \in \mathbb{R}^{q \times q}$ and $\hat{\mathbf{B}}_s^{(k,n)} \in \mathbb{R}^{(d-q) \times (d-q)}$ such that

$$\left\| \begin{pmatrix} \hat{\mathbf{B}}_{\eta}^{(k,n)} & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{B}}_s^{(k,n)} \end{pmatrix} \hat{\Omega}_{1,k,n}(\mathbf{x}) - \mathbf{A}^{-1} \right\| = O_{\mathbb{P}}(r_{k,n}).$$

Other assumptions

- **Finite dependence:** Let $P_1, P_2 \subset \mathcal{U} \times \mathcal{T}$. There exists a fixed $L > 0$ not depending on the sets P_1 and P_2 such that if $\min \{ \|\mathbf{p} - \mathbf{p}'\| : \mathbf{p} \in P_1, \mathbf{p}' \in P_2 \} > L$ then the sets $\{\mathbf{z}(\mathbf{p}) : \mathbf{p} \in P_1\}$ and $\{\mathbf{z}(\mathbf{p}) : \mathbf{p} \in P_2\}$ are independent.
- : It holds that $\sup_{i \in [n_0]} \max_{j \in [d]} \mathbb{E} |z_j(\mathbf{p}_i)|^4 \leq C < \infty$ uniformly for all n_0 .

The main result

We have

$$r_{k,n} = O_{\mathbb{P}} \left(\frac{1}{n} + \frac{1}{kn} \right)$$

The main result

We have

$$r_{k,n} = O_{\mathbb{P}} \left(\frac{1}{n} + \frac{1}{kn} \right)$$

Messages:

- For $\sqrt{n_0}$ -consistent estimation you need $k/n = O(1)$.

Link to slides: [https://github.com/perej1/presentations/
tree/main/nonstationary-field](https://github.com/perej1/presentations/tree/main/nonstationary-field)

References

- Bachoc, François, Marc G Genton, et al. (2020). “Spatial blind source separation”. In: *Biometrika* 107.3, pp. 627–646. DOI: 10.1093/biomet/asz079.
- Bachoc, François, Christoph Muehlmann, et al. (2024). “Large-sample properties of non-stationary source separation for Gaussian signals”. In: *Electronic journal of statistics* 18.1.